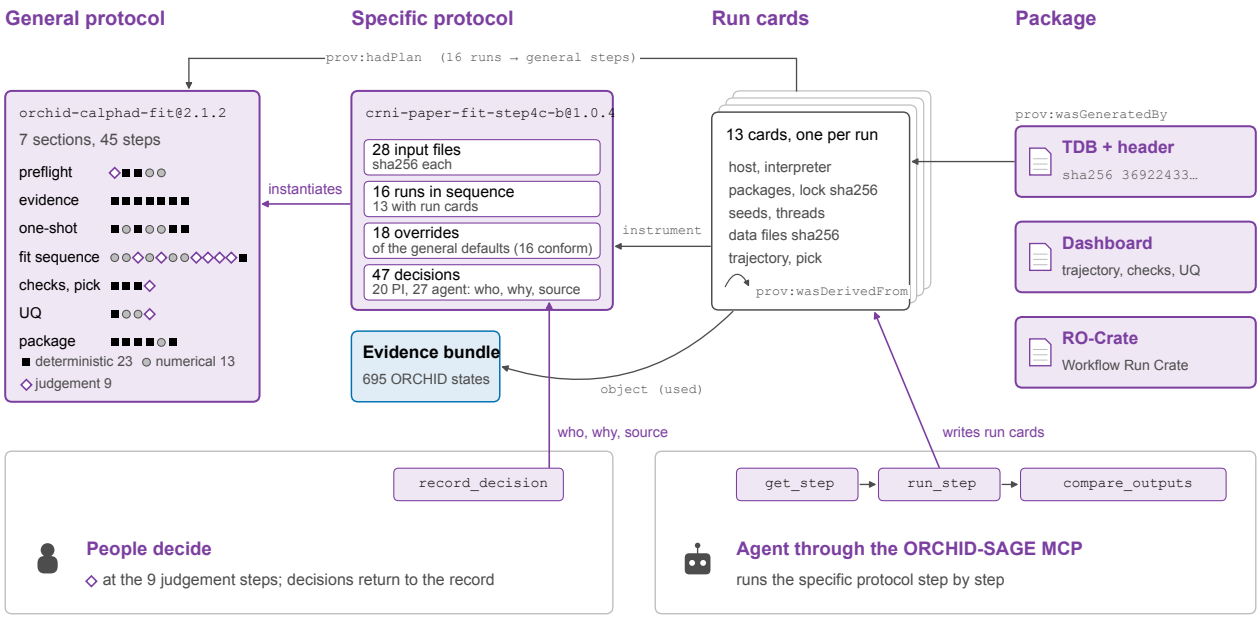

General protocol
Specific protocol
Run cards
Package
prov:hadPlan (16 runs → general steps)
orchid-calphad-fit@2.1.2
7 sections, 45 steps
preflight
evidence
one-shot
fit sequence
checks, pick
UQ
package
■ deterministic 23 ● numerical 13
◇ judgement 9
instantiates
crni-paper-fit-step4c-b@1.0.4
28 input files
sha256 each
16 runs in sequence
13 with run cards
18 overrides
of the general defaults (16 conform)
47 decisions
20 PI, 27 agent: who, why, source
instrument
13 cards, one per run
host, interpreter
packages, lock sha256
seeds, threads
data files sha256
trajectory, pick
prov:wasDerivedFrom
prov:wasGeneratedBy
TDB + header
sha256 36922433…
Dashboard
trajectory, checks, UQ
RO-Crate
Workflow Run Crate
Evidence bundle
695 ORCHID states
object (used)
who, why, source
writes run cards
record_decision
get_step
run_step
compare_outputs
People decide
◇ at the 9 judgement steps; decisions return to the record
Agent through the ORCHID-SAGE MCP
runs the specific protocol step by step